

# Research Direction Proposal

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## Topic Area and Angle

This project focuses on evaluating trust in LLM-based data agents for statistical tasks, specifically in the context of missing data imputation.

Recent work has highlighted the growing role of data agents in automating data science workflows (David Donoho, 2017; Sun Ming et al., 2025). However, much of this work evaluates systems based on task performance, often overlooking whether these systems can correctly interpret statistical assumptions.

This raises an important gap:

Current evaluations tend to measure output correctness, but not whether AI systems can detect incorrect assumptions provided by users.

This project is motivated by the question:

How can we assess trust in human–AI collaboration in statistical tasks via the agent’s ability to recognize incorrect assumptions?

To explore this, I focus on a concrete and common statistical setting: missing data imputation under MNAR (Missing Not At Random).

## Format and Approach

This project will take the form of a small empirical study.

The key idea is to evaluate a data agent based on a comparison design using a concrete dataset with MNAR (Missing Not At Random) missingness. A real-world dataset will be obtained and missingness will be artificially induced to create a controlled MNAR mechanism.

The same dataset will then be presented to the AI system under four different prompt conditions:

- Blind condition: no background information is provided
- Reasoning condition: additional sociological or contextual background is provided to allow inference about the missingness mechanism
- Direct instruction condition: the prompt explicitly states that the data is MNAR

- Misleading condition: the prompt falsely describes the data as randomly missing (e.g., MCAR), simulating human error

Rather than focusing only on the correctness of imputed values, the evaluation emphasizes how the agent responds to different assumptions. Specifically, the analysis will examine:

- Whether the agent passively accepts the given assumption
- Whether it can infer or question the missingness mechanism
- Whether it adjusts its imputation strategy accordingly

This design will allow for a more comprehensive comparison of statistical reasoning behaviour of the agent across different levels of guidance and potential misdirection.

## **Feasibility Concerns**

The main feasibility concern lies in constructing an appropriate evaluation dataset.

In particular, for the reasoning condition, the MNAR structure needs to be discernible from contextual / “common-sense” background but not so obvious that the task becomes trivial. It remains unclear whether to start from a complete dataset and manually introduce missingness, and how complex the data should be to make it both realistic and interpretable.

A second challenge concerns how to evaluate the agent’s statistical performance in a principled way.

It is not yet clear whether evaluation should focus on:

- the correctness of final imputed values
- the statistical procedures used (e.g., whether the agent considers tests such as Little’s MCAR test)
- or the reasoning process reflected in the generated code and explanations

Capturing genuinely sound statistical reasoning and superficially plausible outputs may require a quantitative and qualitative combination.

## **Seed References**

- Donoho (2017): Highlights the central role of statistical thinking in data science, emphasizing that data analysis is not only about computation but also about understanding assumptions. This motivates the focus on assumption-awareness rather than output correctness.
- Sun et al. (2025, Survey on LLM-based Agents): Provides an overview of how large language model agents are used in statistical and data science tasks. The survey reveals that current evaluations largely focus on task performance, leaving a gap in assessing deeper statistical reasoning.

- StatEval (2025): Introduces a benchmark for evaluating statistical reasoning in LLMs and identifies significant limitations in their ability to handle statistical concepts. This directly motivates the need for alternative evaluation criteria, such as assumption recognition.
- DSAEval (2026): Proposes a comprehensive framework for evaluating data science agents across real-world tasks, including multi-level capability assessment (e.g., L0–L5). This inspires the use of a level-based evaluation framework in this project.
- LLMs for Missing Data Imputation (2026): Investigates how LLMs behave in missing data tasks, including hallucination effects and inconsistencies in reasoning. This supports the idea that correct outputs do not necessarily imply correct statistical understanding.